

Sabeeha Syed

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WORK EXPERIENCE

Big Data Engineer - Resonate, Remote

Jun'24 - Aug'25

Led the Data Append build, with a team of 4, from architecture to launch, which is now live and used by enterprise clients

- Designed and deployed **Data Append**, a fully automated system matching customer-provided HEMs, MAIDs, IPs, and addresses against **250M+** US consumer profiles, delivering domain-specific insights from a **15K+ attribute catalog**
- Built multiple **AWS Step Functions** orchestrating pipelines integrating **Snowflake (Snowpipe, Snowpark, Stored Procedures, RBAC)**, **DBT** with custom macros, **Lambdas (Java/Python)**, supporting **S3, Snowflake, and Box.com**
- Used **Databricks, EMR Serverless**, with **Apache Spark** and **Scala** for optimizing **multi-TB** transformations
- Implemented **CI/CD** with **Git** and **Docker**, using **Terraform** for infrastructure-as-code and versioned deployments
- Built streaming and batch data pipelines using **Kafka**, integrating logs and metrics with the **ELK (Elasticsearch, Logstash, Kibana)** for real-time monitoring, troubleshootings and custom **Python scripts** to automate QA validations
- Facilitated all **scrum** ceremonies and partnered with the PMs to align data delivery with **Business Intelligence** needs.

Lead Data Engineer - Georgia Tech, Atlanta

Aug'22 - Jun'24

- Spearheaded a project under **Dr. Feryal Ozel** to build a cloud data lake and pipelines for weather data from **9 EHT telescopes**, supporting **15 teams** across **8 countries** with diverse output formats.
- Developed an **end-to-end data pipeline** using **DBT, Airflow**, and **Redshift**, processing 25+ years of 46GB historical and live **VLBI** data, implementing advanced relational mapping and warehousing with downstream **OLAP**.
- Built an **AWS-based ingestion** system using **API Gateway, Lambda (Boto3), S3, Kinesis Firehose, and Step Functions, Snowflake**, boosting real-time analytics by 20% and handling 5M+ daily data points.
- Automated ingestion with event-driven triggers via **S3 Events, SNS, IAM, SQS, Lambda, and CloudWatch**, ensuring real-time data availability and fault tolerance.
- Designed a scalable **Snowflake-to-GCP** migration system with **BigQuery, Cloud Composer, Pub/Sub, IAM, and Cloud Functions**, enabling secure cross-cloud data mobility.

Data Science Intern - Digital Insomnia, Atlanta, GA

May'23 - Aug'23

- Enhanced sales by 6% weekly through channel analysis, deploying advanced ML models such as **XGBoost, Random Forest, and SVM (Support Vector Machines)** for highly accurate audience prediction and targeting.
- Improved Facebook ad targeting by 12% using a **ChatGPT API** for audience segmentation with **A/B testing**.
- Automated **ETL pipelines in Python (AWS Lambda)** for **Facebook Ad API** data, visualized in **Tableau**
- Audited ad system architecture for clients, driving a **20% increase in click rates** through actionable insights.

Data Engineer - PricewaterhouseCoopers (PWC), Bengaluru, India

Jun'20 - Aug'22

Executed DE and DS initiatives; deployed predictive models, optimized big data transformations for Fortune 500 clients

- Reduced data processing time by 16% by automating and optimizing big data transformation, handling over 2TB monthly via **PySpark** ETL operations, and loading to **SQL** databases from **AWS S3**.
- Applied **NLTK** for **NLP** tasks for tokenization and stemming, while implementing **Word2Vec** and **SVM**. Achieved a notable 9% accuracy enhancement in sentiment analysis for customer service calls.
- Revised data integrity to 99.5% and cut discrepancies by 30% by integrating **Google Pubsub** for efficient ingestion and processing of over 3M transactions daily, alongside rigorous **ETL** validation and quality assurance.

EDUCATION

Georgia Institute of Technology, Atlanta

GPA 3.8

Master's in Computer Science (Spl. in Machine Learning)

Aug'22 - May'24

Vellore Institute of Technology, Vellore, India

GPA 3.6

Bachelor's in Information Technology

June'16 - May'20

PROJECTS

• Append RAI | LangChain, GPT-4o, FAISS, Streamlit, Python, Snowflake, python-pptx

Built an AI Copilot to automate analysis and reporting reducing manual reporting effort by 95%. Designed a RAG pipeline (LangChain + FAISS + GPT-4o) to extract insights from snowflake tables. Delivered PII-aware profiling, and auto-generated PDF/PPT reports. Enabled self-serve analytics via Streamlit, eliminating SQL dependency.

• SCM Insights Automation Engine | Python, UiPath, Snowflake, dbt, Tableau, AWS

Automated data ingestion from SAP using UiPath, and built ETL pipelines in Snowflake to power real-time dashboards. Cut reporting latency by 90%, improved inventory decisions by 25%, and reduced out-of-stock incidents by 12%.

• Financial Fraud Detection | Python, Scikit-Learn, TensorFlow, Power BI

Developed a financial fraud detection system using Random Forest and a multi-layer neural network with three hidden layers (ReLU activation). Processed 6GB of data, achieving 98% accuracy in identifying fraudulent transactions. Visualized fraud patterns and trends with Power BI, providing actionable insights to mitigate risk

SKILLS

Programming: Scala, Python, SQL, Java, NoSQL, R, Excel, VBA, C, C++, Alteryx, Django, Matlab

Visualization: Tableau, Power BI, Google Studio, Plotly, Excel Charts, Pandas, Scikit-Learn, Matplotlib

Database and Cloud: MySQL, PySpark, AWS (EMR, Step Functions, Redshift, RDS, S3, Lambda, API Gateway, Glue, Kinesis, Firehose, IAM), Oracle, Azure, DataBricks, GCP, DynamoDB, Kafka, MongoDB, Agile, JIRA, Confluence

Data Engineering: ETL Processes, Data Architecture, Data Structures, Data Modeling, DBT (Data Build Tool), Airflow